Sahaj Bhadja

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Summary

Systems-oriented software engineer with 2+ years building performant, reliable, privacy-conscious components using **C++, Java, and Python**; experienced in OS fundamentals, concurrency, caching, algorithms, cross-platform services, testing, and debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed performant backend components using Python and FastAPI, processing 5,000+ monthly workflows with concurrent execution, explicit state transitions, failure isolation, error handling, and 99%+ production reliability.
- Optimized indexed data access and service interfaces for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through algorithm analysis, query tuning, validation, and targeted debugging.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Implemented multithreaded Python services with secure file ingestion and nonblocking batch processing, handling 1,000+ files while maintaining stable APIs across 10,000+ daily requests and peak workloads.
- Developed and validated reusable service components from requirements through rigorous testing and technical documentation, resolving defects and reducing feedback turnaround 60% while sustaining 90%+ consistent system behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed efficient algorithms and complex data structures for trajectory planning across 100,000+ records, improving long-horizon performance 18% through Python-based implementation, time-series processing, and reproducible system-level simulation.
- Designed a spatial-temporal architecture integrating road geometry, lane direction, and vehicle state, validating 10,000 scenarios and improving overall closed-loop stability 15% through performance analysis and debugging.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built modular cross-platform data-sharing services using Java, Python, and object-oriented patterns, producing reliable executable builds and 20+ comprehensive automated tests across macOS, Linux, and Windows virtual machines.
- Improved API performance with indexing, field filtering, pagination, and intelligent caching, reducing response times 30%, payload sizes 25%, and technical debt 20% while debugging distributed integration failures.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Designed a secure, privacy-conscious distributed system architecture integrating five lattice-based cryptographic primitives, applying careful data-handling principles and fundamental technical tradeoff analysis to reduce modeled quantum-risk exposure 25%.
- Evaluated consensus algorithms through formal benchmark simulations, co-authoring a peer-reviewed technical paper and clearly presenting design decisions, implementation risks, and measured results at IISc Bengaluru through IEEE.

Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL, Shell, Bash
OS Fundamentals: Processes, threads, memory management, CPU scheduling, file systems, shared state, synchronization, locking, concurrency
Systems: Linux, Unix, cross-platform services, caching, REST APIs, microservices, distributed systems, virtual machines
Engineering: Data structures, algorithms, time complexity, space complexity, object-oriented design, design patterns, interfaces, privacy by design
Quality: Unit testing, integration testing, system testing, validation, debugging, performance profiling, Git, CI/CD, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, OS Fundamentals, Concurrency*

- Implemented six C++ scheduling algorithms with process-state handling, interactive simulation, and Gantt visualization, validating 1,000+ combinations while reducing average turnaround time 25% and debugging time 20%.

Chat2SQL Code | *Java, Cross-Platform Service, Caching*

- Extended a Java and TypeScript service across Linux, macOS, and Windows, implementing schema exploration, query validation, optimized data access, local debugging, and Docker-based delivery for 10+ databases.

Blockchain Authentication Analysis | *Privacy, Security, Distributed Systems*

- Designed a privacy-preserving authentication model for three access scenarios, using decentralized verification and four permission tiers to reduce repeated key-validation steps 50% while protecting stored user records.